

M^5 Day 5: Exponents

Name: _____

1. Simplify and eliminate any negative exponents. Assume all letters denote positive numbers:

a. $(32x^5y^{-3/2})^{2/5}$

b. $\left(\frac{x^8y^{-4}}{16y^{4/3}}\right)^{-1/4}$

c. $\sqrt{8x^2 + 4x^4}$

d. $\left(\frac{x^{3/2}}{\sqrt{y}}\right)\left(\frac{x^{-2}}{y^3}\right)$

2. State whether these statements are True or False. If False, give one example showing why. All variables represent real numbers.

a. $(ab)^{-400} = a^{-400}b^{-400}$

b. $\sqrt{a^2b^2} = ab$

c. $\frac{xy}{a} = \frac{x}{a} \cdot \frac{y}{a}$

d. $(-a)^2 = -(a^2).$

3. Why is $\sqrt{x^2 + y^2}$ always larger than $\max(|x|, |y|)$?